

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	A	B	D		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,D	A,B	A,C			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	A	B	C	A		
SECTION-II	Q.	1	2	3	4	5	6
	A.	16	400	48	8	2	8

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	D	C	D		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,D	A,B,C,D	A,B,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	D	D	B	D		
SECTION-II	Q.	1	2	3	4	5	6
	A.	4	3	8500	704	4	8

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	D	D	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C	A,B,C,D	A,C,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	D	D	A	B		
SECTION-II	Q.	1	2	3	4	5	6
	A.	288	29	1	7	1	9

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

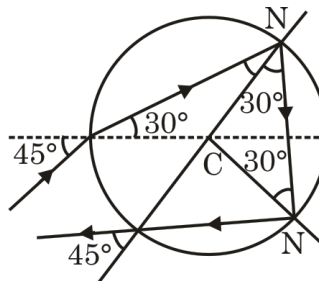
1. Ans (A)

$$\left(A \times 6 \times 0.5 + 8 \times \frac{4}{3} \pi r^3 \right) g = \left(A \times 4 + \frac{4}{3} \pi r^3 \right) \times 1 \times g$$

$$r = \left(\frac{3}{8} \right)^{1/3} \text{ cm}$$

2. Ans (A)

$$\delta = 15^\circ + 120^\circ + 120^\circ + 15^\circ = 270^\circ$$



3. Ans (B)

$$R_0 = \frac{R_{Al} R_{Fe}}{R_{Al} + R_{Fe}}$$

$$R_{Al} = \frac{\rho_{Al} l_{Al}}{A_{al}} = \frac{2.7 \times 10^{-8} \times 50 \times 10^{-3}}{45 \times 10^{-6}}$$

$$= 3 \times 10^{-5} = 30 \times 10^{-6} \Omega$$

$$R_{Fe} = \frac{\rho_{Fe} l_{fe}}{A_{Fe}} = \frac{1.0 \times 10^{-7} \times 50 \times 10^{-3}}{4 \times 10^{-6}}$$

$$= \frac{50 \times 10^{-10}}{4 \times 10^{-6}} = 12.5 \times 10^{-4}$$

$$= 12.5 \times 10^{-4}$$

$$= 1250 \times 10^{-6}$$

$$R_0 = \frac{(30 \times 10^{-6})(1250 \times 10^{-6})}{[30 \times 10^{-6} + 1250 \times 10^{-6}]}$$

$$= \frac{37500 \times 10^{-12}}{1280 \times 10^{-6}} = 29.29 \times 10^{-6}$$

$$= 29.29 \mu\Omega$$

4. Ans (D)

Heat is received by ice from both sides

$$\frac{KA(20)}{0.1} + \frac{K(40)}{0.2} = \frac{Ldm}{dt}$$

$$\frac{dm}{dt} = \frac{KA(400)}{L} = \frac{(336)10^{-4} \times 400}{(80 \times 4.2)1000} \text{ kg/s}$$

$$= 40 \text{ mg/sec}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,B,D)

$$\text{Final pressure} = \frac{kx_0}{S}$$

Workdone by gas = P.E. stored in spring = $\frac{1}{2}kx_0^2$

Change in internal energy

$$\Delta U = -\Delta W = \frac{1}{2} kx_0^2$$

As gas expands, ΔT is negative.

6. Ans (A,B)

$$\text{For } B = \frac{8}{13} \frac{P}{QR}$$

$$\text{Radius of path } r = \frac{mv}{QB} = \frac{p13QR}{Q.8P}$$

$$r = \frac{13}{8}R, \text{ center will beat point C}$$

Equation of circle will be

$$x^2 + \left(y - \frac{5R}{8}\right)^2 = \left(\frac{13}{8}R\right)^2$$

and co-ordinate of point $P_2 \left(\frac{3R}{2}, 0\right)$ will lie on

this circle. (A) is correct.

$$\text{For } B = \frac{2p}{3QR}$$

$$r = \frac{p3QR}{Q.2P} = \frac{3}{2}R$$

$$\text{if } B > \frac{2p}{3QR} \text{ } r \text{ will be } r < \frac{3R}{2}.$$

So particle will re-enter in region (2)

7. Ans (A,C)

Velocity of 3kg block just before collision

$$= \omega \sqrt{a^2 - x^2} = \sqrt{\left(\frac{k}{m}\right)(a^2 - x^2)}$$

$$= \sqrt{\left(\frac{900}{3}\right)(2^2 - 1^2)} = 30 \text{ m/s}$$

Velocity of combined masses immediately after

the collision

$$= \frac{(3)(30)}{3+6} = 10 \text{ m/s}$$

New angular frequency

$$\omega' = \sqrt{\frac{k}{m}} = \sqrt{\frac{900}{9}} = 10$$

$$\text{Therefore } v' = \omega' \sqrt{a'^2 - x^2}$$

$$\Rightarrow 10 = 10 \sqrt{a'^2 - 1^2} \Rightarrow a' = \sqrt{2}m$$

PART-1 : PHYSICS

SECTION-I (iii)

8. **Ans (A)**

$$\text{Displacement } s = \frac{1}{2}at^2 = 16$$

$$W_g = 10(16) = 160 \text{ J}$$

$$W_n = (N \cos \theta) s \quad [\text{Where } N = M(g + a) \cos \theta]$$

$$= \frac{12 \times \sqrt{3}}{2} \times \frac{\sqrt{3}}{2} \times 16 = 144 \text{ J}$$

$$W_f = M(g + a) \sin \theta \cdot s \cos (90 - \theta)$$

$$= 12 \times \frac{1}{4} \times 16 = 48 \text{ J}$$

$$W_{\text{all force}} = \Delta K.E. = \frac{1}{2}mv^2 = \frac{1}{2} \times 1 \times 64 = 32 \text{ J}$$

10. **Ans (C)**

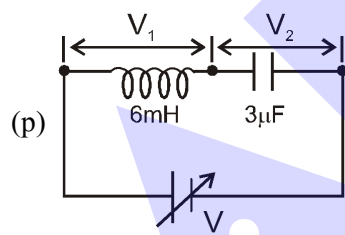
(P) Since $\mu_1 < \mu_2$, the ray of light will bend towards normal after first refraction.

(Q) $\mu_1 > \mu_2$, the ray of light will bend away from the normal after first refraction.

(R) Since $\mu_2 = \mu_3$ means in second refraction there will be no change in the path of ray of light.

(S) Since $\mu_2 > \mu_3$, ray of light will bend away from the normal after second refraction.

11. **Ans (A)**

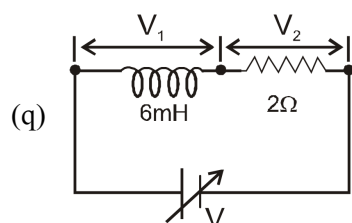


As I is steady state current

$$V_1 = 0; \quad I = 0$$

$$\text{Hence, } V_2 = V$$

So, answer of P $\Rightarrow$ C

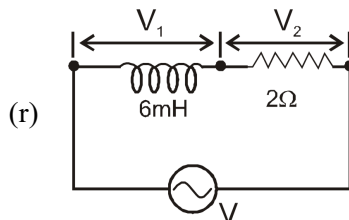


In the steady state ;

$$V_1 = 0 \quad \text{as} \quad \frac{dI}{dt} = 0$$

$$\therefore V_2 = V = IR \quad \text{or} \quad V_2 \propto I \quad \text{and} \quad V_2 > V_1$$

So, answer of q $\Rightarrow$ B, C, D



Inductive reactance $X_L = \omega L$

$$X_L = 6\pi \times 10^{-1} \Omega$$

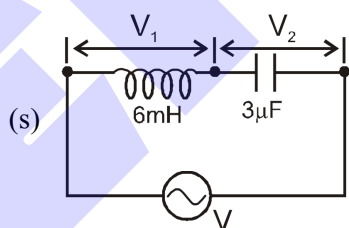
and resistance $= R = 2\Omega$

$$\text{So, } V_1 = IX_L$$

$$\text{and } V_2 = IR$$

$$\text{Hence, } V_2 > V_1$$

So, Answer of r $\Rightarrow$ A, B, D



Here, $V_1 = IX_L$, where, $X_L = 6\pi \times 10^{-1} \Omega$

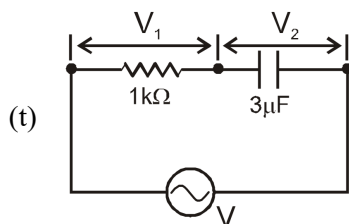
Also, $V_2 = IX_C$, where, $X_C = \frac{10^4}{3\pi}$

$$\text{So, } V_2 > V_1$$

$$V_1 \propto I$$

$$V_2 \propto I$$

So, answer of s $\Rightarrow$ A, B, D



Here, $V_1 = IR$, where,

$$R = 1000 \Omega, \quad X_C = \frac{10^4}{3\pi} \Omega$$

$$V_2 = IX_C, \quad \text{where, } X_C = \frac{10^4}{3\pi}$$

$$\text{So, } V_2 > V_1 \quad \text{and} \quad V_1 \propto I$$

$$V_2 \propto I$$

So, answer of t $\Rightarrow$ A, B, D

Note : For circuit 'p' :

$$V - \frac{L di}{dt} - \frac{q}{C} = 0 \text{ or } CV = CL \frac{di}{dt} + q$$

$$\text{or } 0 = LC \frac{d^2 i}{dt^2} + \frac{dq}{dt} \text{ or } \frac{d^2 i}{dt^2} = -\frac{1}{LC} \frac{dq}{dt}$$

$$\text{So, } i = i_0 \sin \left(\frac{1}{\sqrt{LC}} t + \phi_0 \right)$$

As per given conditions, there will be no steady state in circuit 'p'. So it should not be considered in options of 'c'.

PART-1 : PHYSICS

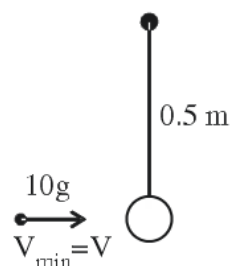
SECTION-II

1. **Ans (16)**

$$20 = (C_1 + C_2) V \Rightarrow V = 2 \text{ volt.}$$

$$Q_2 = C_2 V = 16 \mu C = 16$$

2. **Ans (400)**



$$V' = \sqrt{5gR} = \sqrt{5 \times 10 \times 0.5}$$

$$V' = 5 \text{ m/s}$$

$$m_1 V = m_2 \times 5 - m_1 \times 100$$

$$\frac{10}{1000} \times V = 5 - \frac{10}{1000} \times 100$$

$$V = 400 \text{ m/s}$$

3. **Ans (48)**

The flux passes through ABCD (x - y) plane is zero, because electric field parallel to surface.

Flux of the electric field through surface BCGF (y - z)

At BCGF (electric field)

$$\Rightarrow \vec{E} = 12\hat{i} - (y^2 + 1)\hat{j}$$

$$(x = 3\text{m})$$

$$\text{Flux } \phi_{II} = 12 \times 4 = 48 \text{ Nm}^2/\text{C}$$

$$\text{So } \phi_I - \phi_{II} = 0 - 48 = -48 \text{ Nm}^2/\text{C}$$

$\therefore$ Correct answer -48

4. **Ans (8)**

Sound level decreases by 5dB every km so

sound level decreased in 20 km = 100 dB

$$\beta_2 - \beta_1 = 10 \log_{10} \frac{I_2}{I_1}$$

$$-100 = 10 \log_{10} \frac{I_2}{I_1} \Rightarrow \frac{I_1}{I_2} = 10^{10}$$

$$I_2 = 10^{-10} I_1 \Rightarrow P_2 = 10^{-10} P_1 = 10^{-8} \text{ W}$$

$$x = 8$$

5. **Ans (2)**

Mechanical energy is conserved

$$MgR = M'g \left(\frac{R}{2} \right) + \frac{1}{2} (m') V'^2 + \frac{1}{2} \left[m' \frac{R'^2}{2} \right] \omega'^2$$

$$M' = \frac{M}{4}; \omega' = \frac{V'}{\left(\frac{R}{2} \right)}$$

6. **Ans (8)**

Conserving angular momentum :

$$m(v_1 \cos 60^\circ) 4R = mv_2 R \Rightarrow \frac{v_2}{v_1} = 2$$

Conserving energy of the system :

$$-\frac{GMm}{4R} + \frac{1}{2} mv_1^2 = -\frac{GMm}{R} + \frac{1}{2} mv_2^2$$

$$\Rightarrow \frac{1}{2} v_2^2 - \frac{1}{2} v_1^2 = \frac{3}{4} \frac{GM}{R}$$

$$\Rightarrow v_1^2 = \frac{1}{2} \frac{GM}{R} \Rightarrow v_1 = \frac{1}{\sqrt{2}} \sqrt{64 \times 10^6} = \frac{8000}{\sqrt{2}} \text{ m/s}$$

$$\Rightarrow 500\sqrt{2} X = \frac{8000}{\sqrt{2}} \Rightarrow X = 8$$

PART-2 : CHEMISTRY

SECTION-I (i)

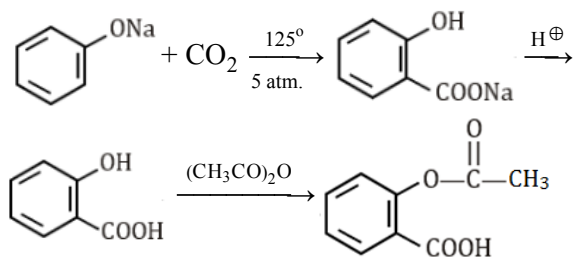
2. **Ans (D)**

$$2.5 \times \frac{2}{5} = \frac{2}{15} \times V \Rightarrow V = 7.5 \text{ ml.}$$

$$K_h = \frac{ch^2}{(1-h)} \Rightarrow 10^{-2} = \frac{0.1 h^2}{(1-h)} \Rightarrow h = 0.27$$

$$[H^+] = ch = 2.7 \times 10^{-2} \text{ M.}$$

3. Ans (C)



PART-2 : CHEMISTRY

SECTION-I (ii)

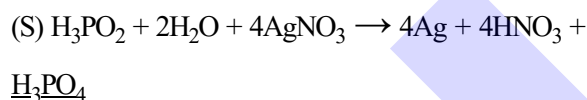
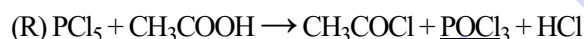
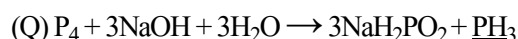
7. Ans (A,B,D)

Because C-1 of both units are linked through glycosidic link in option (C)

PART-2 : CHEMISTRY

SECTION-I (iii)

8. Ans (D)



10. Ans (B)

(P) $x = kt$ (zero order)

$$\frac{1}{3} = k \times 18 \Rightarrow k = \frac{1}{3 \times 18}$$

$$\frac{5}{9} = \frac{1}{3 \times 18} \times t \Rightarrow t = 30 \text{ sec.}$$

(Q) 1st order

$$1 \rightarrow \frac{1/3}{18 \text{ sec}} \rightarrow \frac{2}{3} \rightarrow \frac{1/3}{18 \text{ sec}} \rightarrow \frac{4}{9} \rightarrow \frac{1/3}{18 \text{ sec}} \rightarrow \frac{8}{27}$$

$$1 - \frac{18}{27} = \frac{19}{27}$$

$$\therefore t = t_{19/27} = 54 \text{ sec.}$$

PART-2 : CHEMISTRY

SECTION-II

3. Ans (8500)

$$E_b = E_f + 2RT, A_f = 4A_b$$

$$K = \frac{k_f}{k_b} = 4e^2$$

$$\Delta G^\circ = -RT \ln K = -2500 \ln (4e^2) = -8500 \text{ J mol}^{-1}$$

5. Ans (4)

b, c, e, h

6. Ans (8)

i, ii, iii, iv, vi, vii, viii, x

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)

$$\text{Let } x_i - 5 = d_i$$

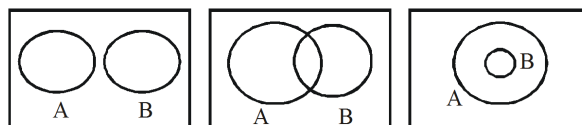
$$\sigma_x^2 = \sigma_d^2 = \frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n} \right)^2$$

$$= \frac{125}{10} - \left(\frac{5}{10} \right)^2 = \frac{25}{2} - \frac{1}{4} = \frac{49}{4}$$

2. Ans (D)

$$\therefore A \not\subseteq B$$

Three possibilities.



only (4) option satisfies all the three possibilities.

3. Ans (D)

Reflexive : (p, q) R(p, q)

$$\Rightarrow pq - qp = 0 \quad (\text{True})$$

Symmetric : (r, s) R(p, q)

$$\Rightarrow rq - sp = 0 \quad (\text{True})$$

Transitive : (p, q) R(r, s) $\Rightarrow$ ps = qr(i)

(r, s) R(t, u) $\Rightarrow$ ru = st(ii)

(i) $\times$ (ii)

$$pu = qt \Rightarrow (p, q) R(t, u) \quad (\text{True})$$

4. Ans (C)

$$2A + B = B^T - A^T \quad \dots(i)$$

take transpose both side

$$2A^T + B^T = B - A \quad \dots(ii)$$

equation (i) – equation (ii)

$$\Rightarrow 3A = -3A^T \Rightarrow A^T = -A$$

$$\Rightarrow 2A + B = B^T + A$$

$$\Rightarrow A + B = B^T$$

$$\Rightarrow |A + B| = |B^T| = |B| = 1$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,C)

$$\begin{aligned} & (5 + 2\sqrt{6})^{2n+1} + (5 - 2\sqrt{6})^{2n+1} \\ &= 2 \left({}^{2n+1}C_0 \cdot 5^{2n+1} + {}^{2n+1}C_2 \cdot 5^{2n-1} \cdot (2\sqrt{6})^2 + \dots \right. \\ & \quad \left. \dots + {}^{2n+1}C_{2n} \cdot 5 \cdot (2\sqrt{6})^{2n} \right) \end{aligned}$$

$$\Rightarrow I + f + f' = 10 \times \text{an integer} \quad \dots(1),$$

where $0 < f' < 1$

also $0 < f < 1$

so $0 < f + f' < 2$

but $f + f'$ must be an integer

$$\Rightarrow f + f' = 1$$

$\therefore (1) \Rightarrow I = \text{a multiple of } 10 - 1 = \text{odd integer}$
and $I + 1 = \text{multiple of } 10$.

Also, the integer next above

$(5 + 2\sqrt{6})^{2n+1}$ is $I + f + f'$, divisible by 10

$$\text{Now } I - 1 = \frac{f}{1-f} = \frac{f}{f'} \Rightarrow If' = f + f'$$

$I \cdot f' = 1$ (not true)

6. Ans (A,B,C,D)

Given that $AA^T = I$ and $BB^T = I$

(A) Let $C = A^2$

$$CC^T = AAA^T A^T = I$$

(B) Let $C = AB$

$$CC^T = ABB^T A^T = I$$

(C) Let $C = AAB B$

$$CC^T = AABBB^T B^T A^T A^T = I$$

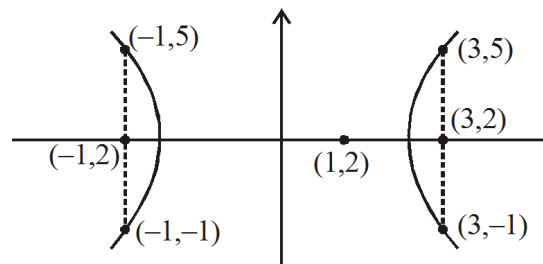
(D) Let $C = A^T B$

$$CC^T = A^T B B^T A = I$$

7. Ans (A,C,D)

$$ae = 2, \frac{b^2}{a} = 3$$

$$a(e^2 - 1) = 3$$



$$\frac{2}{e}(e^2 - 1) = 3$$

$$2e^2 - 3e - 2 = 0$$

$$2e^2 - 4e + e - 2 = 0$$

$$\Rightarrow e = 2 \quad (-1/2 \text{ rejected})$$

foci (3, 2), (-1, 2) $\Rightarrow$ center (1, 2)

$$\frac{a}{e} = \frac{1}{2} \Rightarrow \text{directrix are } x = \frac{3}{2} \text{ and } x = \frac{1}{2}$$

PART-3 : MATHEMATICS

SECTION-I (iii)

8. Ans (D)

(P) Required probability = $\frac{1}{2}$

(Q) Required probability

$$= \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 - \left(\frac{1}{2}\right)^7 = \frac{47}{2^7}$$

(R) Required probability = $\frac{4}{2^7}$

(S) There are five cases 0, 1, 2, 3, or 4 wins

$$\text{Required probability} = \frac{1 + 7 + 15 + 10 + 1}{2^7} = \frac{34}{2^7}$$

9. Ans (D)

$$(P) {}^9C_4 + {}^9C_4 + {}^9C_4 + {}^9C_3$$

$$= {}^{10}C_5 + {}^{10}C_4 = {}^{11}C_5 = {}^{11}C_6$$

$$(Q) = {}^{10}C_5 + {}^{10}C_4 = {}^{11}C_5 = {}^{11}C_6$$

$$(R) {}^9C_5 + {}^9C_5 = 2 \cdot {}^9C_5$$

$$= {}^9C_5 + {}^9C_4 = {}^{10}C_5$$

$$(S) {}^{10}C_5 = 2 \cdot {}^9C_5$$

10. Ans (A)

(P) $z^5 = 1$

$$z = (\cos 2k\pi + i \sin 2k\pi)^{1/5}$$

$$z = \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5}$$

$$k = 0, 1, 2, 3, 4$$

$$\alpha = \cos \frac{2k\pi}{5} \quad \beta = \sin \frac{2k\pi}{5}$$

$$\frac{(\beta^2 - \alpha^2)(\beta^2 + \alpha^2)}{1 - 2\alpha^2} = 1$$

$$\Rightarrow \frac{-\left(\cos\left(\frac{4\pi k}{5}\right)\right)}{-\left(2\cos^2 \frac{2\pi k}{5} - 1\right)} = 1$$

(Q) $z\bar{z} = 1$

$$|2 + \bar{z}|^2 + |2 - z|^2$$

$$(2 + \bar{z})(2 + z) + (2 - z)(2 - \bar{z})$$

$$4 + 2z + 2\bar{z} + 1 + 4 - 2\bar{z} - 2z + 1 = 10$$

(R) $2\alpha\beta = 2(\omega + \omega^2 + \omega^4) \left(\frac{1}{\omega^4} + \frac{1}{\omega^2} + \frac{1}{\omega} \right)$

$$= 2 \left(\frac{1}{\omega^3} + \frac{1}{\omega} + 1 + \frac{1}{\omega^2} + 1 + \omega + 1 + \omega^2 + \omega^3 \right)$$

$$= 2(\omega^4 + \omega^6 + \omega^7 + \omega^5 + 2 + \omega + \omega^2 + \omega^3)$$

$$= 2 \cdot \left(2 + \omega \frac{(\omega^7 - 1)}{\omega - 1} \right) = 4$$

(S) $z_1 + z_3 = z_2 + z_4$

11. Ans (B)

(P) $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a} = -5\vec{a} + 4\vec{b}$

$$\vec{b} \cdot \vec{c} = 5, \vec{a} \cdot \vec{c} = 4 \text{ \& \; } \vec{a} \cdot \vec{b} = 3$$

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = 4\vec{b} - 3\vec{c}$$

(Q) Parametric equation of line is

$$\frac{x - \sqrt{3}}{\cos \theta} = \frac{y}{\sin \theta} = r$$

$$\frac{x - \sqrt{3}}{\cos 60^\circ} = \frac{y}{\sin 60^\circ} = r$$

$$x = \frac{r}{2} + \sqrt{3} \text{ \& \; } y = \frac{\sqrt{3}r}{2}$$

$$\frac{3r^2}{4} = \frac{r}{2} + 2 + \sqrt{3}$$

$$\frac{3r^2}{4} - \frac{r}{2} - (2 + \sqrt{3}) = 0$$

$$r_1 r_2 = \frac{(2 + \sqrt{3})}{3/4} = \frac{4(2 + \sqrt{3})}{3}$$

(R) Point is (4, 2, k)

$$8 - 8 + k = 9$$

$$k = 9$$

(S) $\int_{-\pi/4}^{\pi/4} \frac{1}{\cos^2 x} dx = \int_{-\pi/4}^{\pi/4} \sec^2 x dx = 2[\tan x]_0^{\pi/4} = 2$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (288)

Let B_1 be the boy & G_1, G_2 be the girls who do not want to sit with B_1

× × × × × ×

1 2 3 4 5 6

If B_1 occupies first place then G_1 or G_2 cannot sit on second place

⇒ Number of ways = ${}^4P_2 \times {}^3P_1 = 72$ ways

If B_1 occupies 6th place

similarly number of ways = 72 ways

Now if B_1 sit at 2nd place,

number of ways = ${}^3P_2 \times {}^3P_1 = 36$ ways

Similarly B_1 can sit on 3rd, 4th, 5th place in same number of ways = 36

⇒ Total ways = $72 \times 2 + 36 \times 4 = 288$ ways

2. Ans (29)

Let $B_1, B_2, \dots, B_6$ be six mutually disjoint events where $B_j : j$ appears on the die $1 \leq j \leq 6$.

Clearly $P(B_j) = \frac{1}{6}, 1 \leq j \leq 6$.

Let A denote the event when tail appears exactly four.

$$P(A) = \frac{1}{6} \sum_{j=1}^6 P(A/B_j)$$

$$= \frac{1}{6} \left[0 + 0 + 0 + \frac{{}^4C_4}{2^4} + \frac{{}^5C_4}{2^5} + \frac{{}^6C_4}{2^6} \right]$$

$$P(A) = \frac{29}{384}$$

3. Ans (1)

$$y' = e^{x^2} + 2x^2 e^{x^2}$$

$$y'' = 2xe^{x^2} + 2 \{ 2xe^{x^2} + 2x^3 e^{x^2} \}$$

$$= e^{x^2} \{ 8x + 4x^3 \} = 4xe^{x^2} \{ 2 + x^2 \}$$

$$\Rightarrow y'(0) = 1$$

$$\text{slope} = 1 \quad \frac{-}{0} \quad \frac{+}{0}$$

⇒ $x = 0$ is point of inflection

4. Ans (7)

$$f'(x) = 3px^2 + 2qx + r$$

$$f'(1) = 0 \Rightarrow 3p + 2q + r = 0$$

$$f'(5) = 0 \Rightarrow 75p + 10q + r = 0$$

$$\int_{-1}^1 f(x) dx = 6 \Rightarrow \frac{2q}{3} = 6$$

$$p = -1, r = -15, q = 9$$

5. Ans (1)

$$y^2 + 2yx \frac{dy}{dx} + \frac{y}{1+x^2} + \tan^{-1} x \frac{dy}{dx} = \frac{dy}{dx}$$

$$\Rightarrow \frac{d}{dx}(xy^2) + \frac{d}{dx}(y \tan^{-1} x) = \frac{dy}{dx}$$

$$\Rightarrow xy^2 + y \tan^{-1} x = y + c$$

$$\Rightarrow y(xy + \tan^{-1} x - 1) = c \Rightarrow k = 1$$

6. Ans (9)

$$\alpha^4 - 3\alpha + 1 = 0 \quad \alpha^4 = 3\alpha - 1$$

$$\alpha^3 + \beta^3 + \gamma^3 + \delta^3$$

$$= \frac{1}{\alpha\beta\gamma\delta} [\alpha^4\beta\gamma\delta + \beta^4\alpha\gamma\delta + \gamma^4\alpha\beta\delta + \delta^4\alpha\beta\gamma]$$

$$= \frac{1}{\alpha\beta\gamma\delta} [(3\alpha - 1)\beta\gamma\delta + (3\beta - 1)\alpha\gamma\delta + (3\gamma - 1)\alpha\beta\gamma + (3\delta - 1)\alpha\beta\gamma]$$

$$= \frac{1}{\alpha\beta\gamma\delta} [12\alpha\beta\gamma\delta - \Sigma\alpha\beta\gamma] = \frac{12 - 3}{1} = 9$$